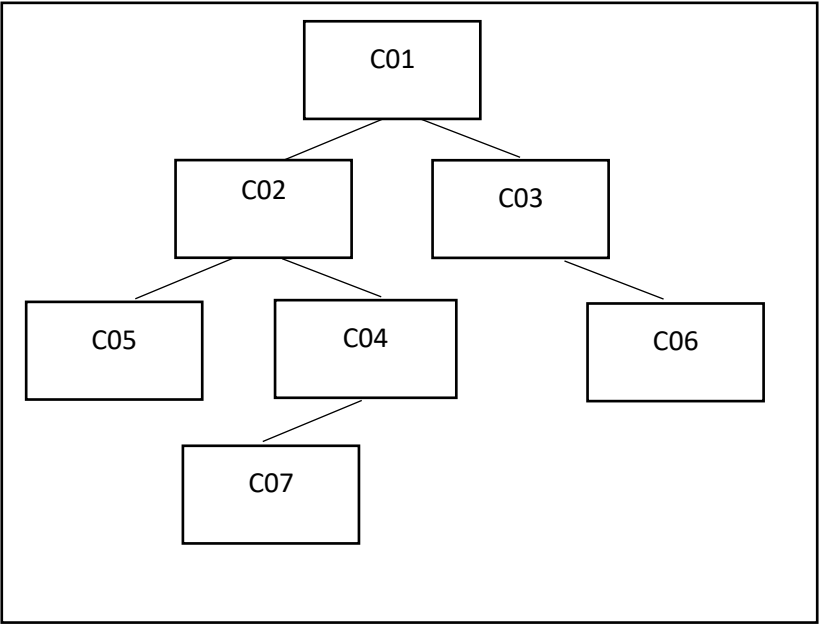


2022 YIJC Mid Year Exam Paper 1 (Theory) – Suggested Solutions

Qn		Solution	Marking Scheme															
1	a i	Use divide by 2 method: $89 \% 2 = 1$ $44 \% 2 = 0$ $22 \% 2 = 0$ $11 \% 2 = 1$ $5 \% 2 = 1$ $2 \% 2 = 0$ $1 \% 2 = 1$ $89_{10} = 101\ 1001_2$	1 1															
	a ii	ASCII code for 'J' is 74_{10} Use divide by 16 method: $74 \% 16 = 10 \rightarrow A$ $4 \% 16 = 4 \rightarrow 4$ $73_{10} = 4A_{16}$ or $0x4A$	1 1															
	b	$2^8 = 128$ Total = $128 + 128 = 256$	1															
	c	<table><tr><td>Hex</td><td>0</td><td>3</td><td>B</td><td>C</td></tr><tr><td>Denary</td><td>0</td><td>3</td><td>11</td><td>12</td></tr><tr><td>Binary</td><td>0000</td><td>0011</td><td>1011</td><td>1100</td></tr></table> $0x03BB = 0000\ 0011\ 1011\ 1100_2$	Hex	0	3	B	C	Denary	0	3	11	12	Binary	0000	0011	1011	1100	1
Hex	0	3	B	C														
Denary	0	3	11	12														
Binary	0000	0011	1011	1100														
	d	Unicode takes up more space when encoded, compared to ASCII ASCII is limited to 127 characters, which is very small compared to Unicode.	1 1															
	e	The 2-byte character is <u>transmitted using UTF-8</u> , encoding the data with <u>3 bytes</u> . <u>Detailed explanation</u> Example: The hexadecimal representation of the Greek character μ is 03BC which is 0000 0011 1011 1100 in binary. Using UTF-8, we will need to use 3 bytes to encode the 16-bits of data: 1110 ----, 10-- ----, 10-- ----. Hence the data is split into: 0000 00 1110 11 1100	1 1															

Qn		Solution	Marking Scheme
		The first byte contains 1110, to indicate that 3 bytes are used to transmit the full data: Byte 1: 1110 0000 The 10 in the next 2 bytes indicate the trailing bytes with data: Byte 2: 1000 1110 Byte 3: 1011 1100 Hence the transmitted data is: 1110 0000 1000 1110 1011 1100	

Qn		Solution	Marking Scheme
2	ai	For 632123, checksum is $6+6+6+4+10+18 = 50$. Modulo result $50 \bmod 11 = 6$. Check digit is 6.	[2]
	aii	For 821321, checksum is $8+4+3+12+10+6 = 43$. Modulo result $43 \bmod 11 = 10$. Check digit is X.	[2]
	b	For 342341X, checksum is $3+8+6+12+20+6 = 55$. Modulo result $55 \bmod 11 = 0$. Check digit is 0, not X. Not a valid ID.	[2] [1]
	c	Length Check (Length must be 7), Format Check (All numbers, with exception of allowing X as the last digit)	[1] [1]

Qn		Solution	Marking Scheme
3	a	 <pre> graph TD C01[C01] --> C02[C02] C01 --> C03[C03] C02 --> C05[C05] C02 --> C04[C04] C04 --> C07[C07] C03 --> C06[C06] </pre>	[1] First 3 in order [1] Left/Right branch is in order [1] All in order

Qn		Solution	Marking Scheme																																
3	b	<table border="1"> <thead> <tr> <th>Index</th><th>Customer ID</th><th>LPtr</th><th>RPtr</th></tr> </thead> <tbody> <tr> <td>0</td><td>C01</td><td>1</td><td>2</td></tr> <tr> <td>1</td><td>C02</td><td>4</td><td>3</td></tr> <tr> <td>2</td><td>C03</td><td>Null</td><td>5</td></tr> <tr> <td>3</td><td>C04</td><td>6</td><td>Null</td></tr> <tr> <td>4</td><td>C05</td><td>Null</td><td>Null</td></tr> <tr> <td>5</td><td>C06</td><td>Null</td><td>Null</td></tr> <tr> <td>6</td><td>C07</td><td>Null</td><td>Null</td></tr> </tbody> </table>	Index	Customer ID	LPtr	RPtr	0	C01	1	2	1	C02	4	3	2	C03	Null	5	3	C04	6	Null	4	C05	Null	Null	5	C06	Null	Null	6	C07	Null	Null	CID correct [1] Ans reflects BST in (a) [2] Mostly Correct [1] All Correct [1]
Index	Customer ID	LPtr	RPtr																																
0	C01	1	2																																
1	C02	4	3																																
2	C03	Null	5																																
3	C04	6	Null																																
4	C05	Null	Null																																
5	C06	Null	Null																																
6	C07	Null	Null																																
	ci	Post order traversal has the priority Left, Right, Self. Starting from the root node C01, visit the left node C02. From this node, visit the left node, C05. As this is a leaf node, it is selected. [C05] Backtrack to C02. Visit the right node c04. Visit and select the node C07. [C07] Backtrack to C04. It is then selected. [C04] Backtrack to C02, where it is selected. [C02] Backtrack to the root, we visit the right node, C03. Then we visit and select [C06] Backtrack to select [C03] Backtrack to finally select the root [C01] The final order is [C05, C07, C04, C02, C06, C03, C01]	[1] Aware of order [1] Visiting, not selecting [1] Backtrack [1] Correct																																
	cii	Using Postal Code 262884, unit number #02-55. Starting at the root, C01's postal code is larger than 262884, therefore we traverse to the left node, index 1. Index 1 contains C02 with postal code 131071, smaller than 262884, therefore we traverse to the right node, index 3. Index 3 contains C04 with unit number #02-55. It is found in the BST.	[1] [1]																																
	ciii	Starting at root node, 615100 is bigger than C01's postal code 524287, so we traverse to the right node, index 2. Index 2 contains C03 with postal code 768675, which is bigger, so we traverse to the left node, but it is null. Therefore, it does not exist.	[1] [1]																																

Qn		Solution	Marking Scheme
4	a	The primary key is <u>LoanID</u>, but not all <u>non-key attributes are fully dependent</u> on the primary key. Title and Author are dependent on BookID	1 1

Qn		Solution	Marking Scheme																																			
		StudentName is dependent on StudentID																																				
	b	The table should be in 2NF. The table should not have transitive dependencies.	1 1																																			
	c	Book<table><tr><th>BookID</th><th>Title</th><th>Author</th></tr><tr><td>2132</td><td>Around The World in 80 Days</td><td>Jules Verne</td></tr><tr><td>1078</td><td>Khmer History</td><td>Tom Smith</td></tr></table> Student<table><tr><th>StudentID</th><th>StudentName</th></tr><tr><td>101</td><td>Chea</td></tr><tr><td>102</td><td>Sohka</td></tr></table> Loan<table><tr><th>LoanID</th><th>BookID</th><th>StudentID</th><th>LoanDate</th><th>ReturDate</th></tr><tr><td>01</td><td>2132</td><td>101</td><td>2 Mar 2022</td><td>12 Mar 2022</td></tr><tr><td>02</td><td>1078</td><td>101</td><td>2 Mar 2022</td><td>5 Mar 2022</td></tr><tr><td>03</td><td>2132</td><td>102</td><td>13 Mar 2022</td><td></td></tr></table> 1 mark for (each table and field names and entries) x 3	BookID	Title	Author	2132	Around The World in 80 Days	Jules Verne	1078	Khmer History	Tom Smith	StudentID	StudentName	101	Chea	102	Sohka	LoanID	BookID	StudentID	LoanDate	ReturDate	01	2132	101	2 Mar 2022	12 Mar 2022	02	1078	101	2 Mar 2022	5 Mar 2022	03	2132	102	13 Mar 2022		1 1 1
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02	1078	101	2 Mar 2022	5 Mar 2022																																		
03	2132	102	13 Mar 2022																																			
	d	Book (<u>BookID</u> , Title, Author) Student (<u>StudentID</u> , StudentName) Loan (<u>LoanID</u> , <u>BookID</u> , <u>StudentID</u> , LoanDate, ReturnDate)	1 1 1																																			
	e	Student Loan Book 1 mark for all entities 1 mark for all relationships	1 1																																			
	f	NoSQL uses dynamic schema, compared to SQL's fixed schema, making it easier to store both structured and unstructured data. NoSQL allows for easy updating to schemas and fields without affecting existing data and structure.	1 1																																			

Qn		Solution	Marking Scheme
5	a i	Affordance is a property or feature which presents a prompt on what can be done with this object. [any relevant example related to webpage UI design]	1 1
	a ii	A mental model is what the user believes about the system. It is built in his brain and referenced when trying to use the system.	1

Qn		Solution	Marking Scheme
		[any relevant example related to webpage UI design]	1
	b	• Unnecessary for user to enter name since login is via email address. Remove it • Too many items in dropdown list for user to browse and select. Consider implementing search • If 4 items can be borrowed, a checkbox form control can be used to allow users to choose • Loan date, since it is fixed to today's date, need not be keyed. Or more useful to show date of return • Typing 3 other titles in a normal text entry form control is tedious, and format may not be correct. Checkbox form control • Cancel button is too prominent, compared to submit link. Switch around • Warning about loan period and disciplinary action are too small and greyed. To move to a more prominent location (perhaps below loan/return date) and made larger. For each issue raised, relevancy 1 mark, improvement 1 mark Max 3 issues. Total 6 marks	

Qn		Solution	Marking Scheme
6	(a)	[2] Advantages: - access rights enforced - centralise management of data, files and backup [1] Disadvantages: - single point of failure when the server or network switch fails. - more expensive to set up and maintain a server-client LAN	
	(b)	- data sent in small units known as packets [1] - transmitted over the 'best route' [1] - acknowledgement by receiver upon receiving each packet [1] - sender will resend the packet(s) not acknowledge by the receiver [1] - receiver to assemble all the packets back to the original data [1]	
	(c)	- TCP: Establish a communication channel [1] - TCP: Break down a message into packets for transmission and reassemble back [1] - IP: Defines how to address and route the packets to the destination [1]	
	(d)	- user type www.amazon.com in the web browser and enter - the request for www.amazon.com is routed to a DNS [1] - the IP address for the web server is return to the computer [1]	

Qn		Solution	Marking Scheme
		- the computer sends the request to the web server using the ip address [1] - the web server reply with the webpage	
	(e)	[Sender Side]: - The sender uses a hash algorithm to create a hashed version of the message[1] - The sender uses his private key to encrypt the hash to the digital signature[1] - Both the message and the digital signature are sent to the receiver[1] [Receiver Side]: - the receive uses the sender's public key issue by the Certificate Authority (CA) [1] - to decrypt the digital signature back to the sender's version of hash; this verifies the authenticity of the sender.[1] - The receiver uses the hash algorithm to create a new hash from the received message[1] If the two hashes match, it means the data is not altered and is sent by the known sender.[1]	

Qn		Solution	Marking Scheme
7	(a)	[1] Validation : To check for compliance to a particular format. [1] Verification: To ensure the data entered is the same as the actual data	
	(b)	[1] Validation: format check (userid@yijc.edu.sg) [1] Verification: double entry for the password	
	(c)	[1] The submitted username and password are recorded in the resulting URL when using HTTP GET method, whereas they are stored separately when using HTTP POST method. [1] As such, the data may remain in the web browser history, be bookmarked or cached, hence it is an insecure way of handling sensitive data.	
	(d)	[1] The new password is hashed and stored in the database server. [1] When a user tries to login, the password is hashed and compared with the hashed password in the database.	

Qn		Solution	Marking Scheme
8	a i)	Personal data refers to data about an individual who can be identified from that data, or from that data and other information to which the organisation has or is likely to have access. Personal data can consist of Full Names, Residential address, NRIC numbers, Mobile telephone numbers, Photographs, personal email address.	[1] [1] with examples
	a ii)	1. Accountability Obligation 2. Notification Obligation 3. Consent Obligation 4. Purpose Limitation Obligation 5. Accuracy Obligation 6. Protection Obligation 7. Retention Limitation Obligation 8. Transfer Limitation Obligation 9. Access and Correction Obligation 10. Data Breach Notification Obligation 11. Data Portability Obligation*	Any 2 [2]
	b	Breach of Protection Obligation: for failing to put in place reasonable security arrangements to prevent the unauthorised disclosure of personal data of applicants (names and emails) to all recipients Breach of Consent Obligation: disclosing the personal data of past applicant without their consent (distributing a previously filled form to assist an applicant to fill up her form) Breach of Retention Limitation Obligation: failure to cease retention of historical sensitive data (or anonymise data) as soon as it is reasonable to assume that the purpose for which that personal data was collected is no longer necessary for business purposes.	[2] [2] (also accepted)
	c	Bob should: 1. Check that names and emails are not revealed to all recipients. Bob should place email addresses in the 'bcc' fields to send emails, to avoid disclosing names and emails to all other recipients. Alternatively, Bob can send emails individually. 2. Create sample forms with fictitious data for reference instead of using previously filled forms as examples to applicants facing difficulties filling up the form.	[1] [1]